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Knife lock

Abstract

A folding knife has a frame and a blade pivotable relative to the frame about a fold axis between an open position and a closed position. A toggle coupled to the frame and pivotable relative to the frame about a toggle axis is configured for selectively locking the blade in the open position. A cantilever spring has a free end configured to selectively retain the toggle in a selected position.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7748122	12/2009	Duey	30/159	B26B 1/048
10695920	12/2019	Bruhns	N/A	B26B 1/044
11123884	12/2020	Mollick	N/A	B26B 1/048
11820028	12/2022	Makari	N/A	B26B 1/044
2003/0070299	12/2002	Frazer	30/160	B26B 1/048
2009/0277015	12/2008	Duey	30/158	B26B 1/046
2015/0239134	12/2014	Duey	30/159	B26B 1/06
2021/0197410	12/2020	Busse	N/A	B26B 1/042
2022/0410415	12/2021	Douglas	N/A	B26B 1/048

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation-in-Part of U.S. Ser. No. 17/410,979, filed 24 Aug. 2021, and claims priority to U.S. 63/069,679, filed 24 Aug. 2020, and to U.S. 63/348,798, filed 3 Jun. 2022, the disclosures of all related applications being incorporated by reference in their entirety.

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is an oblique view of a knife with an embodiment of a lock according to this disclosure.
(2) FIG. **2** is an oblique view of the knife of FIG. **1** with components removed for ease of viewing.
(3) FIG. **3** is a side detail view of the lock of FIG. **1** with a blade of the knife in an open position.
(4) FIG. **4** is a side detail view of the lock of FIG. **1** with a blade of the knife in a closed position.
(5) FIG. **5** is a side detail view of an alternative embodiment of a lock according to this disclosure.
(6) FIG. **6** is a side detail view of another alternative embodiment of a lock according to this disclosure.
(7) FIG. **7** is a side detail view of another alternative embodiment of a lock according to this disclosure.
(8) FIG. **8** is an oblique view of a knife with an embodiment of a toggle retention plate according to this disclosure.
(9) FIG. **9** is an oblique view of the knife of FIG. **8** with components removed for ease of viewing.
(10) FIG. **10** is an enlarged side view of a portion of the knife of FIG. **8**.
(11) FIG. **11** is an oblique view of the lock retention plate of FIG. **8**.
(12) FIG. **12** is an oblique view of a knife with an embodiment of a lock and toggle retention spring

according to this disclosure.
(13) FIG. **13** is an oblique view of an alternative embodiment of a toggle retention plate.

Description

DETAILED DESCRIPTION
(1) In the specification, reference may be made to the spatial relationships between various components and to the spatial orientation of various aspects of components as the devices are depicted in the attached drawings. However, as will be recognized by those skilled in the art after a complete reading of this disclosure, the devices, members, apparatuses, etc. described herein may be positioned in any desired orientation. Thus, the use of terms such as “above,” “below,” “upper,” “lower,” or other like terms to describe a spatial relationship between various components or to describe the spatial orientation of aspects of such components should be understood to describe a relative relationship between the components or a spatial orientation of aspects of such components, respectively, as the device described herein may be oriented in any desired direction.
(2) This disclosure divulges a new type of knife lock that incorporates an extended, secondary lock area, requiring additional motion before the blade is allowed to close.
(3) FIGS. **1** through **4** illustrate a folding knife **11** that incorporates an embodiment of a lock according to this disclosure. Knife **11** comprises a pivotable blade **13**, handles **15**, **17**, and a toggle **19** housed between handles **15**, **17**.
(4) Knife **11** is designed to allow blade **13** to rotate between an open position, shown in FIGS. **1** through **3**, and a closed position, as shown in FIG. **4**. Blade **13** comprises a working portion **21** with sharpened edge **23** and a tang **25**, and blade **13** is pivotable about blade pivot pin **27** on fold axis **29**, pivot pin **27** extending through hole **31** formed in tang **25**. In the closed position, edge **23** is located between handles **15**, **17**. Tang **25** comprises a stop pin surface **33** configured for engaging a stop pin **35** located between handles **15**, **17** and used to prevent further rotation of blade **13** when blade **13** is moved to the open position. Tang **25** also comprises a clearance surface **37**, a primary lock face **39**, radial surface **41**, and a recess **43** for receiving toggle **19** when blade **13** is in the closed position. Recess **43** has stop face **45** on one end and lock face **47** on the opposite end. Primary lock face **39** of tang **25** is located on the opposite side of fold axis **29** from working portion **21**, and the forward end of face **39** is spaced a radial distance from fold axis **29**.
(5) In the embodiment shown, toggle **19** has a figure-8 shape and rotates at a pivot end on pivot pin **49** about toggle axis **50**, which is spaced a radial distance from fold axis **29**. A free end of toggle **19** comprises an open lock face **51**, which is configured to engage primary lock face **39** when blade **13** is in the open position, and a closed lock face **53**, which is configured to engage lock face **47** of tang **25** when blade **13** is in the closed position. A thumb stud **55** extends from at least one side of the free end and through an arcuate slot **57** formed in an adjacent handle **15**, **17**. In some embodiments, thumb studs **55** will extend from opposing sides of toggle **19** to allow for ambidextrous operation. A cap **59** may be installed on an outer end of each thumb stud **55** on an exterior of each handle **15**, **17**.
(6) When blade **13** is in the open position shown in FIG. **3**, stop pin surface **33** engages stop pin **35**, preventing further rotation of blade **13**. Also, the free end of toggle is rotated rearward, so that open lock face **51** engages primary lock face **39**, preventing blade **13** from rotating away from the open position. Toggle **19** is biased by a spring or other device toward the position shown in FIG. **3** and to the rearward extent of slots **57**. Primary lock face **39** may be formed with an angle, as shown, to allow for continued proper locking through additional rearward rotation of toggle **19** after wear occurs on lock face **39** of tang **25** or lock face **51** of toggle **19**.
(7) A secondary lock zone **61** is the angle between a forward end of primary lock face **39** and a plane **52** defined by fold axis **29** of blade **13** and toggle axis **50** of toggle **19**. In preferred

embodiments, the angle of lock zone **61** is between 5 and 35 degrees (inclusive). Secondary lock face **62** is the portion of surface **37** in zone **61** and is configured so that, when the free end of toggle **19** is within zone **61**, lock face **51** of toggle **19** engages secondary lock face **62** for limiting rotation of blade **13** toward the closed position of FIG. **4**. The biasing force on toggle **19** causes toggle **19** to rotate back toward the locked position of FIG. **3** if the user removes closing force applied to blade **13** and/or forward force applied to thumb studs **55**.

(8) To close blade **13**, a user must apply sufficient forward pressure on thumb stud **55** to rotate toggle **19** to where lock face **51** is moved out of zone **61** and to the other side of plane **52**, so that lock face **51** will not engage secondary lock face **62**. This allows blade **13** to be rotated toward the closed position and for the remainder of clearance surface **37** and radial surface **41** to cause the free end of toggle **19** to rotate forward and upward toward the position shown in FIG. **4**. In this closed position, blade **13** is prevented from further rotation by the contact of stop face **45** and the pivot end of toggle **19**. In this embodiment, lock face **53** of toggle **19** engages lock face **47** of tang **25**, preventing blade **13** from rotating away from the closed position and requiring the user to apply upward force to thumb studs **55** to disengage toggle **19** from blade **13**.

(9) Toggle **19** is shown with a figure-8, or “dog bone,” configuration, but toggle **19** may have other shapes, including configurations with linear features. While shown as having thumb studs **55** protruding from toggle **19** and out of handles **15**, **17**, alternative embodiments can include recessed features that do not protrude from handles **15**, **17**, such as, for example, a dished area. Other alternative embodiments incorporate alternative toggle designs, links, levers, and/or similar components for actuating toggle **19**, including versions that allow for reversing the direction of movement for unlocking toggle **19**.

(10) For example, FIG. **5** illustrates an alternative embodiment, in which knife **63** comprises toggle **65**. Toggle **65** is configured similarly to toggle **19**, as described above, although toggle **65** lacks thumb studs **55** and incorporates a thumb lever **67** located on the end of toggle **65** opposite lock face **69**. The outer end of lever **67** protrudes beyond the upper edge of handles **15**,**17** (**17** not shown), both lacking slots **57**. This allows a user to pull rearward on lever **67** to rotate toggle **65** and unlock blade **13**.

(11) FIGS. **6** and **7** illustrate alternative embodiments of the lock according to this disclosure. In FIG. **6**, clearance surface **71** comprises a planar secondary lock face **73** configured to engage lock face **51** when face **51** is within zone **61**. In FIG. **7**, clearance surface **75** is configured like surface **37** but also comprises a female relief **77**, or divot, in secondary lock face **79** that is configured to receive lock face **51** when aligned with relief **77**. Relief **77** provides a positive stop during unintentional rotation of the free end of toggle **19** toward fold axis **29**.

(12) FIGS. **8** through **11** illustrate knife **111**, which is constructed similarly to knife **11**, as described above, and components thereof. Knife **111** comprises a toggle lock like that of knife **11**, and like components have corresponding numbers. In addition, knife **111** also comprises an embodiment of a toggle retention plate **201** according to this disclosure.

(13) Knife **111** comprises a pivotable blade **113**, handles **115**, **117** coupled to frame **118** in this embodiment, and a toggle **119** pivotally carried in frame **118**. Blade **113** is rotatable about blade pivot pin **127**, and stop pin **135** limits rotation of blade **113** when blade **113** is moved to the open position, as shown. Toggle **119** rotates at a pivot end on pivot pin **149**, and a free end of toggle **119** comprises a stud **155** that extends from at least one side of the free end and through an arcuate slot **157** formed in an adjacent handle **115**, **117** and frame **118**. In the embodiment shown, studs **155** extend from opposing sides of toggle **119** to allow for ambidextrous operation. A cap **159** is installed on an outer end of each stud **155** on an exterior of each handle **115**, **117**.

(14) In addition to the friction of the engaged faces of the lock when blade **113** is in the open position, as described above and shown in detail in FIG. **3**, retention plate **201** provides an additional means for retaining toggle **119** in the locked position. Plate **201** has a planar body **203** having an aperture **205** sized for receiving pivot pin **149**. Aperture **207** is formed in body **203** to

allow stud **155** to freely move within aperture **207**, which is shown as an arcuate slot generally corresponding to the shape and length of slot **157**. Aperture **209** is formed in body **203** on the other side of aperture **207** from aperture **205** and is shown as an arcuate slot. A retention spring **211** is formed as a cantilever beam between aperture **209** and aperture **207**, aperture **209** being shown as generally following the curvature of aperture **207** and extending for a least a portion of the length of aperture **207** from near locked-position end **213** of aperture **207**. Aperture **209** intersects aperture **207** to form a free end **215** of spring **211**, with free end **215** having a width that forms a narrowed region in aperture **207**. Aperture **209** may have different curvature than aperture **207**, thereby forming spring **211** in a desired shape and allowing for tuning of the characteristics of spring **211**, such as a spring rate. The motion of each stud **155** is defined by the distance of studs **155** from toggle axis of toggle **119**, so apertures **207**, **209** may have alternative shapes from the arcuate slots shown in the figures. For example, aperture **207** may have a contour that defines a larger area, such as that shown with broken lines **217** in FIG. **11**. Also, apertures **207**, **209** may define a less curved or straight spring.
(15) As best shown in FIG. **10**, retention plate **201** is installed on frame **118** with aperture **207** generally aligned with slot **157**, allowing stud **155** to travel within aperture **207** as toggle **119** is pivoted about pivot pin **149**. As toggle **119** is moved within aperture **207** to the locked position, as shown, stud **155** deforms spring **211** toward aperture **209**, allowing stud **155** to move past the narrowed portion of aperture **207** and into locked-position end **213** of aperture **207**. A portion of free end **215** of spring **211** then engages stud **155** for retaining toggle **119** in the locked position. To release toggle **119** for folding blade **113**, a user must apply sufficient pressure to stud **155** to overcome the bias of spring **211** and deform spring **211**, thereby allowing stud **155** to move in aperture **207** away from locked-position end **213**.
(16) In the embodiment shown, two retention plates **201** are installed, with one located on each side of frame **118**, though other embodiments may have only one plate **201** installed. Also, it should be noted that aperture **209** may be formed on either side of aperture **207**.
(17) Referring to FIG. **12**, knife **301** comprises a lock similar to those described above and an alternative embodiment of toggle retention lock according to this disclosure. Knife **301** comprises a pivotable blade **113** and a toggle pivotally carried in frame **318**. Blade **113** is rotatable relative to frame **318**, and stop pin **135** limits rotation of blade **113** when blade **113** is moved to the open position, as shown. The toggle rotates at a pivot end on pivot pin **149**, and a free end of the toggle comprises a stud **155** that extends from at least one side of the free end and through an arcuate slot **357** formed in frame **318**. Instead of using a separate plate, such as plate **201**, for a toggle retention lock, the lock features are formed in frame **318**. A cantilever spring **311**, which is constructed and operated like spring **211**, is formed in frame **318** and defined by apertures **309**, **357**, which are both shown as arcuate slots but may have alternative configurations. Spring **311** has a free end **315** configured for engaging stud **155** when stud **155** is moved to a locked-position end **313** of aperture **357**.
(18) FIG. **13** illustrates another embodiment of a cantilever spring formed as a separate component for coupling to a knife frame. Spring plate **401** has a planar body **403** having an aperture **409** formed in body **203** for defining spring **411**, aperture **409** being shown as an arcuate slot. Spring **411** is formed as a cantilever beam with free end **415**, which is configured to engage a toggle stud for selectively retaining toggle stud at locked-position opening **413**. Aperture **409** may have different a different shape that that shown, thereby forming spring **411** in a desired shape and allowing for tuning of the characteristics of spring **411**, such as a spring rate, and aperture **409** may define a less curved or straight spring **411**.
(19) It should be noted for each embodiment that the stud may be a separate component coupled to the toggle or be an integral portion of the toggle extending therefrom. Alternatively, toggle may have a recess, dimple, or other feature that the retention spring engages.
(20) At least one embodiment is disclosed, and variations, combinations, and/or modifications of

the embodiment(s) and/or features of the embodiment(s) made by a person having ordinary skill in the art are within the scope of the disclosure. Alternative embodiments that result from combining, integrating, and/or omitting features of the embodiment(s) are also within the scope of the disclosure. Where numerical ranges or limitations are expressly stated, such express ranges or limitations should be understood to include iterative ranges or limitations of like magnitude falling within the expressly stated ranges or limitations (e.g., from about 1 to about 10 includes, 2, 3, 4, etc.; greater than 0.10 includes 0.11, 0.12, 0.13, etc.). For example, whenever a numerical range with a lower limit, R.sub.l, and an upper limit, R.sub.u, is disclosed, any number falling within the range is specifically disclosed. In particular, the following numbers within the range are specifically disclosed: R=R.sub.l+k*(R.sub.u−R.sub.l), wherein k is a variable ranging from 1 percent to 100 percent with a 1 percent increment, i.e., k is 1 percent, 2 percent, 3 percent, 4 percent, 5 percent, . . . 50 percent, 51 percent, 52 percent, . . . , 95 percent, 96 percent, 95 percent, 98 percent, 99 percent, or 100 percent. Moreover, any numerical range defined by two R numbers as defined in the above is also specifically disclosed. Use of the term “optionally” with respect to any element of a claim means that embodiments with and without the element are within the scope of the claim. Use of broader terms such as comprises, includes, and having should be understood to provide support for narrower terms such as consisting of, consisting essentially of, and comprised substantially of. Accordingly, the scope of protection is not limited by the description set out above but is defined by the claims that follow, that scope including all equivalents of the subject matter of the claims. Each and every claim is incorporated as further disclosure into the specification and the claims are embodiment(s) of the present invention. Also, the phrases “at least one of A, B, and C” and “A and/or B and/or C” should each be interpreted to include only A, only B, only C, or any combination of A, B, and C.

Claims

1. A folding knife, comprising: a frame; a blade coupled to the frame and pivotable relative to the frame about a fold axis between an open position and a closed position; a toggle coupled to the frame and pivotable relative to the frame about a toggle axis, the toggle being configured for selectively locking the blade in the open position; and a cantilever spring having a free end configured to selectively retain the toggle in a selected position.
2. The knife of claim 1, further comprising: a stud extending from the toggle; wherein the free end engages the stud for retaining the toggle in the selected position.
3. The knife of claim 1, wherein the spring is formed in the frame.
4. The knife of claim 1, wherein the spring is coupled to the frame.
5. The knife of claim 1, wherein the spring is defined by apertures on each side of the spring.
6. The knife of claim 1, wherein: the blade has a working portion and a tang; the toggle axis is located a radial distance from the fold axis, a free end of the toggle having an open lock face located a radial distance from the toggle axis; the tang has a primary lock face located on the opposite side of the fold axis from the working portion; and when the blade is in the open position, the open lock face engages the primary lock face to prevent rotation of the blade toward the closed position.
7. The knife of claim 6, wherein a forward end of the primary lock face is spaced a radial distance from the fold axis.
8. The knife of claim 6, wherein to allow the blade to pivot to the closed position the free end of the toggle must be rotated through an angle sufficient to move the open lock face from a forward end of the primary lock face to the other side of a plane defined by the fold axis and the toggle axis.
9. The knife of claim 6, further comprising: a clearance surface on the tang and extending from the primary lock face to beyond a plane defined by the fold axis and toggle axis.
10. The knife of claim 6, further comprising: a clearance surface on the tang extending from the

primary lock face to at least a plane defined by the fold axis and toggle axis; wherein a secondary lock face is defined as a portion of the clearance surface between a forward end of the primary lock face and the plane, the secondary lock face being configured to engage the open lock face of the toggle when the open lock face is between the primary lock face and the plane.

11. A folding knife, comprising: a frame; a blade coupled to the frame and pivotable relative to the frame about a fold axis between an open position and a closed position; a toggle coupled to the frame and pivotable relative to the frame about a toggle axis, the toggle being configured for selectively locking the blade in the open position; and a cantilever spring formed in the frame and having a free end configured to selectively retain the toggle in a selected position.

12. The knife of claim 11, further comprising: a stud extending from the toggle; wherein the free end engages the stud for retaining the toggle in the selected position.

13. The knife of claim 11, wherein the spring is defined by at least one aperture formed in the frame.

14. The knife of claim 11, wherein: the blade has a working portion and a tang; the toggle axis is located a radial distance from the fold axis, a free end of the toggle having an open lock face located a radial distance from the toggle axis; the tang has a primary lock face located on the opposite side of the fold axis from the working portion; and when the blade is in the open position, the open lock face engages the primary lock face to prevent rotation of the blade toward the closed position.

15. The knife of claim 11, wherein to allow the blade to pivot to the closed position the free end of the toggle must be rotated through an angle sufficient to move the open lock face from a forward end of the primary lock face to the other side of a plane defined by the fold axis and the toggle axis.

16. A folding knife, comprising: a frame; a blade coupled to the frame and pivotable relative to the frame about a fold axis between an open position and a closed position; a toggle coupled to the frame and pivotable relative to the frame about a toggle axis, the toggle being configured for selectively locking the blade in the open position; and a cantilever spring coupled to the frame and having a free end configured to selectively retain the toggle in a selected position.

17. The knife of claim 16, further comprising: a stud extending from the toggle; wherein the free end engages the stud for retaining the toggle in the selected position.

18. The knife of claim 16, wherein the spring is formed in a plate and defined by at least one aperture formed in the plate.

19. The knife of claim 16, wherein: the blade has a working portion and a tang; the toggle axis is located a radial distance from the fold axis, a free end of the toggle having an open lock face located a radial distance from the toggle axis; the tang has a primary lock face located on the opposite side of the fold axis from the working portion; and when the blade is in the open position, the open lock face engages the primary lock face to prevent rotation of the blade toward the closed position.

20. The knife of claim 16, wherein to allow the blade to pivot to the closed position the free end of the toggle must be rotated through an angle sufficient to move the open lock face from a forward end of the primary lock face to the other side of a plane defined by the fold axis and the toggle axis.